

Generation Interconnection Request System Impact Study Report

For Queue Project Q21-1725

Lone Tree 138 kV
December 30, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1725. The Interconnection Customer proposes a 177.4 MW solar photovoltaic facility interconnecting at the Lone Tree 138 kV substation in the Virginia area. The request was received on April 19, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 470 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$10,009,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Osceola / Michigan Power Lp 138.0 kV	Craven County Wood Energy Lp / Lone Tree Substation 138.0 kV	126.9	571	100.0
Lone Tree / Osceola 138.0 kV	Craven County Wood Energy Lp / Lone Tree 138.0 kV line	126.5	578	100.0
Craven County Wood Energy Lp / Lowmoor Substation 138.0 kV	Osceola / Lone Tree 138.0 kV	114.7	633	100.0
Craven County Wood Energy Lp / Lone Tree 138.0 kV	Lone Tree / Osceola 138.0 kV Line	114.5	639	100.0
Imhoff Tap 1 / Fudge Hollow 138.0 kV	Lowmoor 345 kV/138 kV Transformer	110.6	183	16.8
Lowmoor / Fudge Hollow 138.0 kV	Lowmoor 345/138 kV Transformer	106.2	190	16.8
Ola / Imhoff Tap 1 138.0 kV Line	Lowmoor 345 kV/138 kV Transformer Bank	101.5	181	16.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Osceola / Michigan Power Lp 138.0 kV	Rebuild line with higher-capacity conductor	\$10,628,000	24% (\$2,573,000)
Lone Tree / Osceola 138.0 kV	Rebuild line with higher-capacity conductor	\$4,638,000	24% (\$1,123,000)
Craven County Wood Energy Lp / Lowmoor 138.0 kV Line	Rebuild line with higher-capacity conductor	\$10,433,000	24% (\$2,525,000)
Lone Tree / Craven County Wood Energy Lp 138.0 kV	Rebuild line with higher-capacity conductor	\$8,050,000	24% (\$1,949,000)
Fudge Hollow / Imhoff Tap 1 138.0 kV	Rebuild line with higher-capacity conductor	\$4,282,000	19% (\$832,000)
Lowmoor / Fudge Hollow 138.0 kV	Reconductor line	\$2,676,000	19% (\$520,000)
Imhoff Tap 1 / Ola 138.0 kV	Reconductor line	\$2,506,000	19% (\$487,000)
	Total allocated to project		\$10,009,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.